

SHYAN HAFEEZ KHAN

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Final year Mechanical Engineering student on track for a 1st, with a genuine passion for the automotive sector and high-performance engineering. With an understanding of engineering first principles, I have pursued technically rigorous projects including an electromagnetic variable valve timing prototype and CFD aerofoil analysis, developing strong end-to-end engineering skills across design, simulation, and build. I have demonstrated leadership, having directed a 13-member multidisciplinary team and I enjoy working in fast paced environments as demonstrated by my work for the BT Group undertaken in parallel to my university studies. I am eager to apply my technical foundations and enthusiasm in a challenging graduate or placement role within the automotive or high-performance engineering industry.

RELAVENT PROJECT EXPERIENCE

ADAPTIVE ELECTROMAGNETIC VALVE TIMING (AEVT) PROJECT October 2025 – May 2026

- ❖ Designed and built a solenoid-based cam-less valve actuation prototype as a fully variable alternative to conventional cam-driven systems, achieving up to 15mm of valve lift with spring return.
- ❖ Implemented an Arduino-based control system with MOSFET switching and optocoupler isolation to generate variable-frequency PWM signals simulating engine throttle and speed.
- ❖ Completed end-to-end development covering mechanical design, material selection, circuit design, and system testing, evaluating actuator frequency response and the feasibility of electronically variable valve timing for improved engine efficiency and reduced pumping losses.

MACHINE LEARNING-BASED ADAPTIVE TRAIN SUSPENSION PROJECT January 2026 – April 2026

- ❖ Proposed and simulated an ML-enhanced adaptive electronic suspension system for high-speed rail carriages, formulating a two-degree-of-freedom mechanical model and training an LSTM autoencoder on healthy vibration data to detect track anomalies in real time.
- ❖ Designed a coordinated safety response integrating rapid secondary damping adjustment and automatic speed reduction, shifting suspension behaviour from reactive control to predictive, data-driven fault detection.

CFD PROJECT - NACA0012 AEROFOIL ANALYSIS October 2025 – December 2025

- ❖ Performed a CFD analysis of turbulent flow over a NACA0012 hydrofoil in ANSYS Fluent using the k- ω SST turbulence model, computing lift and drag coefficients across angles of attack from 0° to 20° for tidal turbine blade applications.
- ❖ Conducted a mesh sensitivity study comparing coarse and fine structured quadrilateral grids, validating computational accuracy for both attached and separated flow regimes and confirming aerodynamic stall onset between 16° and 20°.

BIOFUEL STORAGE - AC ADSORPTION MEASUREMENT SYSTEM October 2025 – December 2025

- ❖ Contributed to designing a portable, low-cost system for measuring activated carbon adsorption fill-up behaviour in biogas transport, enabling methane storage testing at pressures up to 12 bar within a £2,000 budget.
- ❖ Produced design sketches and led concept selection, supporting a dual-container pressure vessel arrangement sized to test 200g of AC and transportable over rough terrain by a single person.

BUSINESS DEVELOPMENT PROJECT - WEARABLE CONCUSSION SENSOR March 2025 – June 2025

- ❖ Led a 13-member multidisciplinary team in designing AxonGuard, a wearable concussion sensor for sports headgear integrating force-sensitive resistors, accelerometers, and a low-power radio transmitter for real-time, pitch-wide impact alerting across a dual-format product (helmet and mesh cap).
- ❖ Conducted market research and user surveys validating demand for affordable head impact detection in university sports.
- ❖ Produced a full business case including financial forecasting, supply chain strategy, go-to-market plan targeting university teams and grassroots clubs, and a scalability roadmap with future adaptations for construction and military applications.

AUTONOMOUS TRACK-FOLLOWING ROBOT PROJECT October 2024 – March 2025

- ❖ Designed, built, and programmed an autonomous robot to follow a 15m track and accurately launch two ping pong balls into target zones at different distances, using LDR Wheatstone bridge sensing and PID-tuned motor control.
- ❖ Engineered a servo-actuated elastic band firing mechanism triggered on wall contact, and contributed to CAD modelling, PCB soldering, and structured testing throughout development.

WIND TURBINE AEROFOIL DESIGN PROJECT April 2024 – June 2024

- ❖ Designed, constructed, and tested a wind turbine aerofoil model with the objective of maximising power extraction efficiency, applying blade aerodynamics and structural design principles.
- ❖ Analysed performance across multiple fan output levels by measuring current, voltage, and power to compute the coefficient of performance (Cp) and tip speed ratio, identifying optimal operating conditions and sources of experimental error.

IMECHE FOUNDATION DESIGN CHALLENGE – AUTONOMOUS VEHICLE October 2023 – December 2023

- ❖ Designed and built an autonomous vehicle for the 2024 IMechE Foundation Design Challenge, capable of travelling to a vertical wall and returning to its starting point via a DPDT switch-based motor direction reversal circuit.
- ❖ Calculated gear reduction ratios and motor specifications to meet target run time and repeatability requirements within a constrained £35 budget.

SUSTAIBILITY PROJECT - HYDROPONIC PLANT GROWING SYSTEM

June 2019 – May 2020

- ❖ Conceptualised and developed an indoor hydroponic plant growing system to promote sustainable urban gardening, conducting research into urban agriculture techniques and analysing user needs and environmental constraints to inform the design.
- ❖ Produced multiple design iterations using sketches and CAD software, then prototyped the system across various materials, iterating with peer feedback to ensure functionality, durability, and aesthetic quality.

WORK EXPERIENCE

ENGINEERING STUDENT AMBASSADOR,

December 2024 – June 2026

FACULTY OF SCIENCE AND TECHNOLOGY, LANCASTER UNIVERSITY (Lancaster, United Kingdom)

- ❖ Inspired prospective students by representing the Engineering Department at open days and outreach events.
- ❖ Demonstrated the capabilities and credibility of the department showing off state of the art equipment and use cases.
- ❖ Delivered campus tours and department presentations, effectively communicating course content, facilities, and student life.
- ❖ Contributed to widening participation efforts by mentoring applicants from underrepresented backgrounds.

TELECOMMUNICATIONS SALES,

November 2024 – June 2025

EE (BT GROUP) (Lancaster, United Kingdom)

- ❖ Consistently exceeded sales targets, such as the mobile targets (up by 12%) and the customer enrichment targets (up by 15%).
- ❖ Studied the product range and common customer demographics to support informed sales conversations.
- ❖ Delivered tailored solutions and upsold services, increasing retention and average revenue per user (ARPU) by up to 10%.
- ❖ Managed high-volume interactions with professionalism, resolving billing, upgrade, and contract queries efficiently.
- ❖ Retained customers at risk of leaving by identifying needs and offering personalised retention packages.
- ❖ Built rapport with a diverse range of customers, leading to improved sentiment scores (17%) and reduced complaint rates.
- ❖ Coached colleagues to share best practices and improve customer service and sales techniques.

FINANCIAL ADMINISTRATOR,

November 2017 – November 2018

EXCEL ENERGY SOLUTIONS LTD (London, United Kingdom)

- ❖ Maintained accurate financial records by systematically recording day-to-day transactions and reconciling accounts.
- ❖ Organised physical and electronic receipts by date from purchases made from the business accounts.
- ❖ Displayed transactions from these receipts onto Microsoft Excel spreadsheets to calculate and deduct VAT from business-related purchases using categories for the types of expenses.

EDUCATION

LANCASTER UNIVERSITY (Lancaster, United Kingdom)

October 2023 – Present

Bachelors in Mechanical Engineering - Expecting a 1:1.

RR6 SIXTH FORM, RUTLISH SCHOOL (London, United Kingdom)

September 2020 – August 2022

A-Levels – Mathematics, Further Mathematics, Physics, Chemistry.

Senior Maths Challenge 2021 – Awarded Silver (Top 16%).

RUTLISH SCHOOL (London, United Kingdom)

January 2016 – August 2020

GCSEs – Grades 9-7 In 11 Subjects including English Language and Maths. FSMQ Level 3 Additional Maths Qualification (A).

Intermediate Maths Challenge 2018 – Awarded Bronze. Junior Maths Challenge 2017 – Awarded Bronze (Top 25%).

University of Arts London Level 2 Drawing Award at the UAL National Saturday Club: South.

Duke of Edinburgh Bronze Award.

VOLUNTEERING EXPERIENCE

CHARITY HIKES

November 2023 – April 2026

- ❖ Participated in charity hikes across locations including the Lake District, Peak District, and East Sussex, raising funds for various charitable causes such water wells and food in drought-stricken countries and war-torn through self-managed fundraising campaigns on social media.
- ❖ Demonstrated commitment to community impact and physical endurance by consistently volunteering time outside of academic and professional commitments.

FIRST GIVE CHARITY PROGRAMME

June 2018

RUTLISH SCHOOL (London, United Kingdom)

- ❖ Spearheaded a successful fundraising campaign, leading a team of classmates to raise £2,775 for Scope, a disability charity.
- ❖ Executed fundraising events such as school-wide cake sales, exceeding the initial fundraising target of £570 by 480%.
- ❖ Delivered a persuasive presentation to a panel of judges, peers, and parents, articulating the campaign's journey and achievements, which resulted in winning the £1,000 First Give grand prize.
- ❖ Demonstrated strong leadership, teamwork, and communication skills throughout the project, motivating the team and engaging with the wider community to support the cause.